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BSCS 1-Y1-1

ACTIVITY1

- **Problem Solving(Complete this activity using [1]- if else statement and [2]-switch case statement):**

[1]- if else statement

Syntax:

```
import java.util.Scanner;
```

```
public class StudentGradingIfElse {  
    public static void main(String[] args) {  
        Scanner input = new Scanner(System.in);  
  
        String choice;  
  
        do {  
  
            System.out.print("Java Programming Score: ");  
            double javaScore = input.nextDouble();  
  
            System.out.print("C Programming Score: ");  
            double cScore = input.nextDouble();  
  
            System.out.print("Database Handling Score: ");  
            double databaseScore = input.nextDouble();  
  
            // average
```

```
double average = (javaScore + cScore + databaseScore) / 3;

if (average >= 90 && average <= 100) {
    System.out.println("Grade: A");
}
else if (average >= 80 && average <= 89) {
    System.out.println("Grade: B");
}
else if (average >= 75 && average <= 79) {
    System.out.println("Grade: C");
}
else if (average < 75) {
    System.out.println("Grade: F");
}
else {
    System.out.println("Invalid score.");
}

System.out.printf("Average: %.3f\n", average);

// Ask if user wants to continue
System.out.print("Do you want to continue: YES / NO: ");
choice = input.next();

} while (choice.equalsIgnoreCase("YES"));

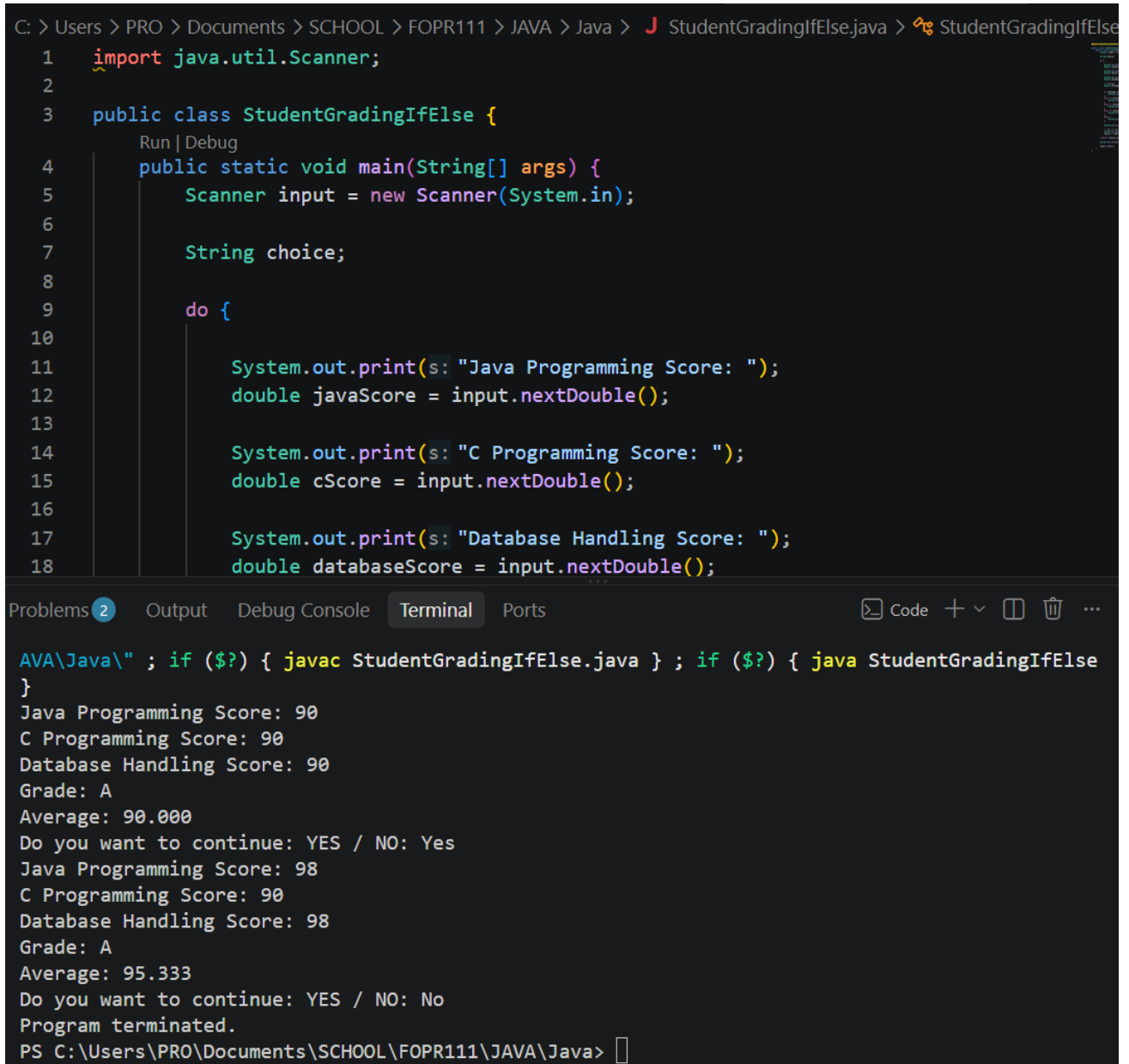
System.out.println("Program terminated.");
```

```

        input.close();
    }
}

```

Output:



The screenshot shows an IDE with a Java file named `StudentGradingIfElse.java`. The code defines a `main` method that prompts the user for scores in Java, C, and Database Handling, calculates the average, and determines a grade based on the average. It also includes a loop to ask if the user wants to continue.

```

1  import java.util.Scanner;
2
3  public class StudentGradingIfElse {
4      public static void main(String[] args) {
5          Scanner input = new Scanner(System.in);
6
7          String choice;
8
9          do {
10
11             System.out.print(s: "Java Programming Score: ");
12             double javaScore = input.nextDouble();
13
14             System.out.print(s: "C Programming Score: ");
15             double cScore = input.nextDouble();
16
17             System.out.print(s: "Database Handling Score: ");
18             double databaseScore = input.nextDouble();

```

The terminal output shows the program's execution for two test cases:

```

AVA\Java\" ; if ($?) { javac StudentGradingIfElse.java } ; if ($?) { java StudentGradingIfElse
}
Java Programming Score: 90
C Programming Score: 90
Database Handling Score: 90
Grade: A
Average: 90.000
Do you want to continue: YES / NO: Yes
Java Programming Score: 98
C Programming Score: 90
Database Handling Score: 98
Grade: A
Average: 95.333
Do you want to continue: YES / NO: No
Program terminated.
PS C:\Users\PRO\Documents\SCHOOL\FOPR111\JAVA\Java>

```

```

1  import java.util.Scanner;
2
3  public class StudentGradingIfElse {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner input = new Scanner(System.in);
7
8          String choice;
9
10         do {
11
12             System.out.print(s: "Java Programming Score: ");
13             double javaScore = input.nextDouble();
14
15             System.out.print(s: "C Programming Score: ");
16             double cScore = input.nextDouble();
17
18             System.out.print(s: "Database Handling Score: ");
19             double databaseScore = input.nextDouble();
20
21             // average
22             double average = (javaScore + cScore + databaseScore) / 3;
23
24             if (average >= 90 && average <= 100) {
25                 System.out.println(x: "Grade: A");
26             }
27             else if (average >= 80 && average <= 89) {
28                 System.out.println(x: "Grade: B");
29             }
30             else if (average >= 75 && average <= 79) {
31                 System.out.println(x: "Grade: C");
32             }
33             else if (average < 75) {
34                 System.out.println(x: "Grade: F");
35             }
36             else {
37                 System.out.println(x: "Invalid score.");
38             }
39
40             System.out.printf(format: "Average: %.3f%n", average);
41
42             // Ask if user wants to continue
43             System.out.print(s: "Do you want to continue: YES / NO: ");
44             choice = input.next();
45
46         } while (choice.equalsIgnoreCase(anotherString: "YES"));
47
48         System.out.println(x: "Program terminated.");
49
50         input.close();
51     }
52 }

```

[2]-switch case statement

Syntax:

```
import java.util.Scanner;
```

```
public class StudentGradingSwitch {  
    public static void main(String[] args) {  
        Scanner input = new Scanner(System.in);  
  
        String choice;  
  
        do {  
            // Input scores  
            System.out.print("Java Programming Score: ");  
            double javaScore = input.nextDouble();  
  
            System.out.print("C Programming Score: ");  
            double cScore = input.nextDouble();  
  
            System.out.print("Database Handling Score: ");  
            double databaseScore = input.nextDouble();  
  
            double average = (javaScore + cScore + databaseScore) / 3;  
  
            int category;
```

```
if (average >= 90 && average <= 100) {
    category = 1;
}
else if (average >= 80) {
    category = 2;
}
else if (average >= 75) {
    category = 3;
}
else if (average < 75) {
    category = 4;
}
else {
    category = 0;
}

// Switch-case for grade
switch (category) {
    case 1:
        System.out.println("Grade: A");
        break;

    case 2:
        System.out.println("Grade: B");
        break;

    case 3:
```

```
        System.out.println("Grade: C");
        break;

    case 4:
        System.out.println("Grade: F");
        break;

    default:
        System.out.println("Invalid score.");
    }

    System.out.printf("Average: %.3f%n", average);

    // Ask if user wants to continue
    System.out.print("Do you want to continue: YES / NO: ");
    choice = input.next();

    } while (choice.equalsIgnoreCase("YES"));

    System.out.println("Program terminated.");

    input.close();
}
}
```

Output:

```
1  import java.util.Scanner;
2
3  public class StudentGradingSwitch {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner input = new Scanner(System.in);
7
8          String choice;
9
10         do {
11             // Input scores
12             System.out.print(s: "Java Programming Score: ");
13             double javaScore = input.nextDouble();
14
15             System.out.print(s: "C Programming Score: ");
16             double cScore = input.nextDouble();
17
18             System.out.print(s: "Database Handling Score: ");
19             double databaseScore = input.nextDouble();
20
21             double average = (javaScore + cScore + databaseScore) / 3;
22
23             int category;
24
25             if (average >= 90 && average <= 100) {
26                 category = 1;
27             }
28         } while (choice != "n");
29     }
30 }
```

Problems 2 Output Debug Console Terminal Ports Code + -

```
PS C:\Users\PRO\Documents\SCHOOL\FOPR111\JAVA\Java> cd "c:\Users\PRO\Documents\SCHOOL\FOPR111\JAVA\Java\" ; if ($?) {
tudentGradingSwitch.java } ; if ($?) { java StudentGradingSwitch }
Java Programming Score: 90
C Programming Score: 90
Database Handling Score: 90
Grade: A
Average: 90.000
Do you want to continue: YES / NO: YES
Java Programming Score: 88
C Programming Score: 78
Database Handling Score: 98
Grade: B
Average: 88.000
Do you want to continue: YES / NO: NO
Program terminated.
PS C:\Users\PRO\Documents\SCHOOL\FOPR111\JAVA\Java>
```

```
C:\Users\PRO > PRO > Documents > SCHOOL > FOPR111 > JAVA > Java > J StudentGradingSwitch.java > StudentGradingSwitch > main(String[])
1  import java.util.Scanner;
2
3  public class StudentGradingSwitch {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner input = new Scanner(System.in);
7
8          String choice;
9
10         do {
11             // Input scores
12             System.out.print(s: "Java Programming Score: ");
13             double javaScore = input.nextDouble();
14
15             System.out.print(s: "C Programming Score: ");
16             double cScore = input.nextDouble();
17
18             System.out.print(s: "Database Handling Score: ");
19             double databaseScore = input.nextDouble();
20
21             double average = (javaScore + cScore + databaseScore) / 3;
22
23
24             int category;
25
26             if (average >= 90 && average <= 100) {
27                 category = 1;
28             }
29             else if (average >= 80) {
30                 category = 2;
31             }
32             else if (average >= 75) {
33                 category = 3;
34             }
35             else if (average < 75) {
36                 category = 4;
37             }
38             else {
39                 category = 0;
40             }
41
42             // Switch-case for grade
43             switch (category) {
44                 case 1:
45                     System.out.println(x: "Grade: A");
46                     break;
47
48                 case 2:
49                     System.out.println(x: "Grade: B");
50                     break;
51
52                 case 3:
53                     System.out.println(x: "Grade: C");
54                     break;
55
56                 case 4:
57                     System.out.println(x: "Grade: F");
58                     break;
59
60                 default:
61                     System.out.println(x: "Invalid score.");
62             }
63
64             System.out.printf(format: "Average: %.3f\n", average);
65
66             // Ask if user wants to continue
67             System.out.print(s: "Do you want to continue: YES / NO: ");
68             choice = input.next();
69
70         } while (choice.equalsIgnoreCase(anotherString: "YES"));
71
72         System.out.println(x: "Program terminated.");
73
74         input.close();
75     }
76 }
```

ACTIVITY 2:

Write a java Program to perform the following Arithmetic Operations :

- Addition, 2)Subtraction, 3)Multiplication, 4)Division, 5)Modulus, 6)Increment, 6)Decrement

NOTE: The user will submit two values

Syntax:

```
import java.util.Scanner;
```

```
public class ArithmeticOperations {  
    public static void main(String[] args) {  
  
        Scanner input = new Scanner(System.in);  
  
        double x, y, result = 0.0;  
  
        System.out.print("Enter value of x: ");  
        x = input.nextDouble();  
  
        System.out.print("Enter value of y: ");  
        y = input.nextDouble();  
  
        System.out.println("\nVariable values:");  
        System.out.println("x = " + x);  
        System.out.println("y = " + y);  
    }  
}
```

```
System.out.println("result = " + result);

// Arithmetic Operations
System.out.println("\nArithmetic Operation:");

result = x + y;
System.out.println("Addition: x + y = " + result);

result = x - y;
System.out.println("Subtraction: x - y = " + result);

result = x * y;
System.out.println("Multiplication: x * y = " + result);

result = x / y;
System.out.println("Division: x / y = " + result);

result = x % y;
System.out.println("Modulus: x % y = " + result);

result = x++;
System.out.println("Increment: x++ = " + x);

result = x--;
System.out.println("Decrement: x-- = " + (x - 1));

input.close();
```

```
}  
}
```

Output:

```
1  import java.util.Scanner;  
2  
3  public class ArithmeticOperations {  
4      Run | Debug  
5      public static void main(String[] args) {  
6  
7          Scanner input = new Scanner(System.in);  
8  
9          double x, y, result = 0.0;  
10  
11         System.out.print(s: "Enter value of x: ");  
12         x = input.nextDouble();  
13  
14         System.out.print(s: "Enter value of y: ");  
15         y = input.nextDouble();  
16  
17         System.out.println(x: "\nVariable values:");  
18         System.out.println("x = " + x);  
19         System.out.println("y = " + y);  
20         System.out.println("result = " + result);  
21  
22         // Arithmetic Operations  
23         System.out.println(x: "\nArithmetic Operation:");  
24  
25         result = x + y;  
26         System.out.println("Addition: x + y = " + result);  
27  
28         result = x - y;  
29         System.out.println("Subtraction: x - y = " + result);  
30  
31         result = x * y;  
32         System.out.println("Multiplication: x * y = " + result);  
33  
34         result = x / y;  
35         System.out.println("Division: x / y = " + result);  
36  
37         result = x % y;  
38         System.out.println("Modulus: x % y = " + result);  
39  
40         result = x++;  
41         System.out.println("Increment: x++ = " + x);  
42  
43         result = x--;  
44         System.out.println("Decrement: x-- = " + (x - 1));  
45  
46         input.close();  
47     }  
48 }
```

Problems 2 Output Debug Console Terminal Ports

PS C:\Users\PRO\Documents\SCHOOL\FOPR111\JAVA\Java> cd "c:\Users\PRO\Documents\SCHOOL\FOPR111\JAVA\Java" ; if (\$?) {
rithmeticOperations.java } ; if (\$?) { java ArithmeticOperations }
Enter value of x: 10
Enter value of y: 10

Variable values:
x = 10.0
y = 10.0
result = 0.0

Arithmetic Operation:
Addition: x + y = 20.0
Subtraction: x - y = 0.0
Multiplication: x * y = 100.0
Division: x / y = 1.0
Modulus: x % y = 0.0
Increment: x++ = 11.0
Decrement: x-- = 9.0
PS C:\Users\PRO\Documents\SCHOOL\FOPR111\JAVA\Java> }